

Digitalization in agriculture. Towards an integrative approach

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ABSTRACT

Farm systems around the world have become less complex, dominated by monocultures, but at the same time much more complicated due to the increasing demands from society, leading to ever increasing compliance and regulatory requirements. Digital technologies have the potential to help alleviate such complications by virtue of their capacity to handle large quantities of information and by automating tasks. However, the opposite seems to be happening. The continuous proliferation of technologies being marketed to farmers adds to the complexity of farming. Farmers are feeling overwhelmed by the abundance of alternative technologies and are complaining about a lack of compatibility between them. We took the view that all the digital elements on farm need to be considered as a system, rather than looking at each component in isolation. We call such system the “digital layer” of a farm. Taking an engineering approach, we started by cataloguing the tasks that a generic farmer would routinely perform. We described 178 farm tasks, including strategic (11), operational (130) and reporting (37) tasks. This catalogue was organized in the form of an ontology, and guided us to develop a systematic description of the functional requirements of the digital layer. The inventory of functional requirements is a critical element in the design of the digital layer. We conducted a network analysis considering the data interdependencies between tasks, and observed a relatively high level of interconnectivity between tasks. This explains why digital technologies developed in isolation are struggling with interoperability issues, and, at least in part the reasons for farmers’ frustrations. This led us to conclude that the farm digital layer needs to be holistically designed, delivered, and supported, to support farm tasks coherently. This would require a much higher degree of coordination than we currently observe in the agri-tech sector, and possibly the need for central coordination entities.

1. Introduction

Farming around the world has been getting increasingly more scrutinised because of increasing demands around food security and environmental sustainability. Apart from the intrinsic difficulties of farming, such as weather and market fluctuations, there are now increasing demands from society for it to become more sustainable and accountable (de Olde et al., 2018; Fischer, 2018). These lead to ever-increasing requirements from consumers, regulators, processors, retailers and other downstream actors in the supply chains. Where these supply chains cross national borders, the requirements increase even further. In addition, digital technologies set to enter the agricultural sector in the coming years have enormous disruptive potential (Trendov et al., 2019), and they come with mixed blessings (Espig et al., 2022; Jakku et al., 2023). On the one hand, some more techno-optimistic views see these technologies as opportunities to tackle the challenges of modern-day farming

(Mondejar et al., 2021). But on the other hand, the continuous proliferation of technologies being marketed to farmers adds another layer of complexity to farming. At the coalface, farmers are feeling overwhelmed by the abundance of alternative technologies and complaining about a lack of compatibility between them, resulting in issues such as data double handling (Romera et al., 2024). In summary, the situation appears to be quite messy, and this is the problem that our research concentrates on.

Digital technologies are becoming increasingly common on farms, but the full potential of digitalization has yet to be realized (Trendov et al., 2019). For example, a recent survey in New Zealand indicate that 41 % of farmers do not see much value in digital technologies, and that the willingness to adopt them is lower than outside the primary sector (AgriTech New Zealand, 2022). Eastwood et al. (2023) conducted an extensive literature review on the barriers to digitalization in agriculture and identified four categories: “technology design”, “farm systems and

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supply chains,” “agricultural innovation system and policy,” and “societal”. In our study, we concentrate on the first category identified by Eastwood et al., that is technology design. This category encompasses several key barriers to digitalization in agriculture, such as poor data and technological interoperability, a lack of confidence in the technologies, poor value proposition, and a lack of fit with specific farm contexts. We hypothesise that many of the barriers on the other categories may in part emanate from here.

The present study approaches the problem from an engineering design perspective in an effort to bring structure to the issue at hand. To accomplish this, we follow the process detailed by Pahl et al. (2007), which outlines a series of structured steps in the planning and design process (Fig. 1). The first step in this process, “clarification of the task,” involves gathering information on the requirements and constraints of the product, and what are the functions it is meant to fulfil. To do so, we view the full set of digital elements on a farm as our ‘system to be designed’, which we refer to as “the digital layer”. The overarching purpose of such digital system is to support farming tasks in a multitude of ways. Shepherd et al. (2020) argue that the transformational potential of digitalization in agriculture lies in the integration of a broad range of technologies, including sensors, telecommunication, data storage, analytics, robotics, and more. The digital layer encompasses all that. It is related to the “platform approach” proposed by Amiri-Zarandi et al. (2022), which they describe as “a unified solution that facilitates cooperation in smart farming applications”. While these authors focused on data processing and sharing, our starting point was a systematic description of the functions that digital technologies could fulfil within farm systems. Amiri-Zarandi et al. (2022) proposed six performance requirements for the design of their “smart farming platforms”: 1) covers basic farm functional needs, 2) flexibility for integration/interoperable, 3) flexibility for Scale / modularity, 4) security and privacy / access, 5) reliable capacity for real-time and 6) compliance with changing regulations. These requirements resonate with the FAIR data principles (Findable, Accessible, Interoperable, Reusable, cf. Wilkinson et al., 2016). Our contributions consists of a detailed analysis of **functional requirements** for a farm system, including the development of a preliminary ontology of farm tasks. Furthermore, scholars have used the term “research by design” to refer to the way in which design processes generate new knowledge (Le Masson et al., 2009; Sevaldson, 2022). By adopting this research by design view, we aim to better understand the reasons for the persistent barriers to farm digitalization and to propose potential solutions. For that, we follow a structured planning and design process, as shown in Fig. 1 (adapted from Pahl et al., 2007).

2. Methods

We first developed a hierarchy of farm tasks, and then we commissioned an experienced farm consultant, and ex-farm manager, to identify tasks within each category in the hierarchy, for a generic dairy farm in New Zealand. The consultant also was asked to describe each in terms of inputs, outputs, controls, and means (ISO, 2012):

- **Function** (or task in our case): A transformation of inputs to outputs, by means of some mechanisms, and subject to certain controls (e.g. milking cows).
- **Inputs**: that which is transformed by a function (or task in our case) into output(s) (e.g., cows to be milked).
- **Outputs**: that which is produced by a function (or task in our case) (e.g., milked cows and milk).
- **Controls**: a condition or set of conditions required for a function to produce correct output(s) (e.g., milking shed protocols)
- **Means**: the means used by a function (or task in our case) to transform input into output(s) (e.g., milking system, personnel).

The information provided by the consultant was utilized as the basis to create an ontology of farm tasks. This was done using Protégé, “a free,

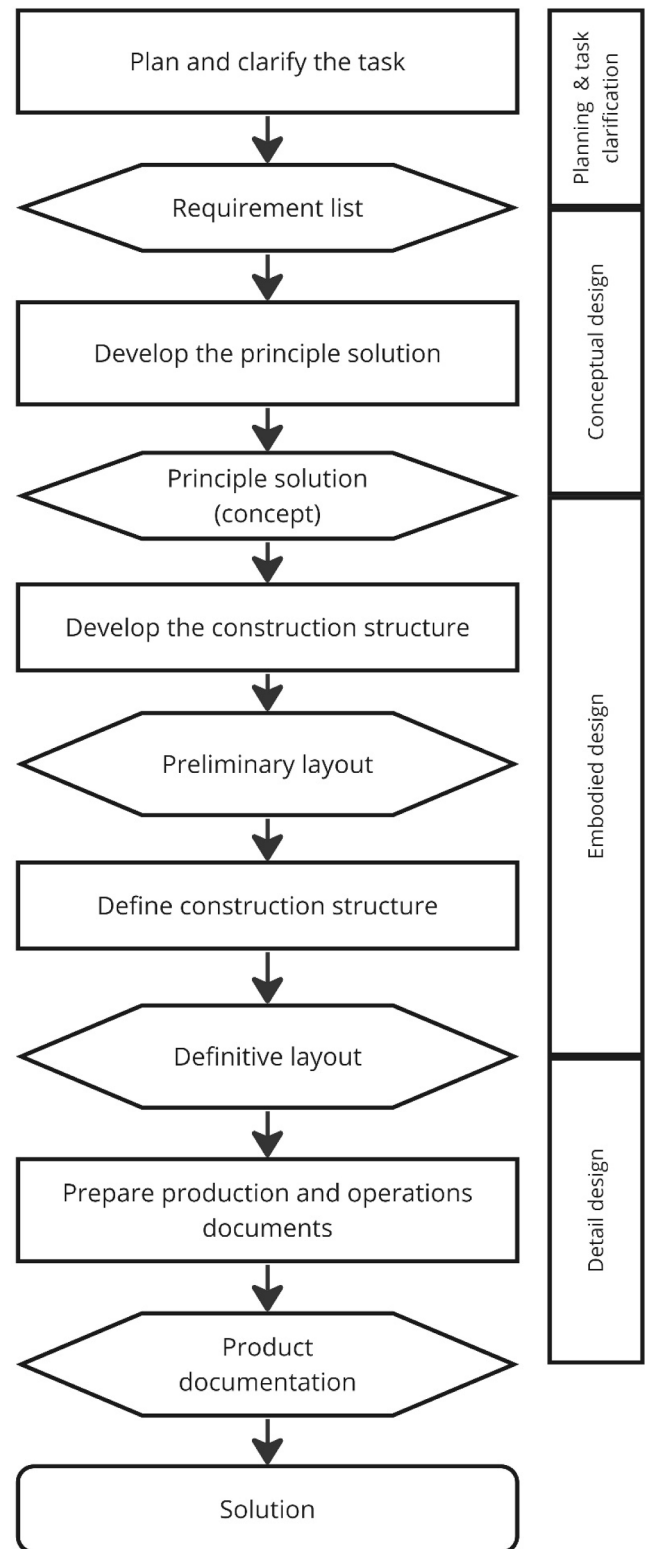


Fig. 1. Steps in the planning and design process (). adapted from Pahl et al., 2007

open-source ontology editor and framework for building intelligent systems”, developed by the Stanford Center for Biomedical Informatics Research at the Stanford University School of Medicine (Musen, 2015). Ontologies are particularly useful to characterize systems components, but more importantly to understand the rich relationships between them. This is an essential element of systemic design approaches (Sevaldson, 2022).

Once the ontology was completed, we were able to query (using SPARQL query syntax in Protégé) to extract all the data items that were used or produced (see Fig. 2) for each of the tasks. This allowed us to analyse dependencies between tasks (i.e., if they share data). For that, we employed network analysis techniques, using a free software called Visone (Brandes and Wagner, 2004). Visone “is a software for the visual creation, transformation, exploration, analysis, and representation of network data, jointly developed at the University of Konstanz and the Karlsruhe Institute of Technology since 2001” (Visone wiki).

3. Results

3.1. Task catalogue

The Class hierarchy for the Class “Task” is presented in Table 1.

3.2. Ontology

It is important to indicate that the ontology is not comprehensive yet as more tasks could be identified or described in more detail. However, we considered that it was on a stage advanced enough for our purpose. Furthermore, the ontology can now be extended at any time.

The ontology contains nine main classes (Task, DataItem, Control, Responsible, Means, Action, Input and Output). These are interrelated as depicted in Fig. 2. The ontology permits multiple inheritance, for example, and individual of the class DataItem can also be an Input or an Output of a task individual.

The ontology contains a total of 661 individuals, categorized into 174 Classes, Table 2 presents more details. The main classes can be briefly described as:

- **Task:** A type of activity (or function) that needs to be performed for the functioning of the farm (see Table 1 for the list of subclasses in the ontology).
- **DataItem:** a piece of information about any entity of the farm. It also includes complex information, which contains other multiple data-items, such as farm management policies.
- **Control:** something that regulates the realization of a task, e.g. regulatory requirements, internal management policies or industry standards.
- **Responsible:** person responsible for the execution of a task.
- **Means:** tools and mechanisms required for the realization of a task, e.g., a piece of equipment or infrastructure.
- **Action:** the specific action executed by a task, e.g., milking cows, culling animals, or conserve forage.
- **Input:** materials or information used (not necessarily consumed) for the execution of the task.
- **Output:** materials or information produced in the execution of the task.

The full description of the ontology can be viewed in <https://doi.org/10.6084/m9.figshare.23722101>. This file includes five sheets:

1. All Tasks: this includes, for each task individual:
 - a. The class hierarchy
 - b. TaskNr: Unique ID of the tasks.
 - c. taskName: Name of the tasks.
 - d. comment: Description of the tasks.
 - e. executes: actions executed by each task.
 - f. produces: outputs produced by each task.
 - g. uses: inputs used by each task.

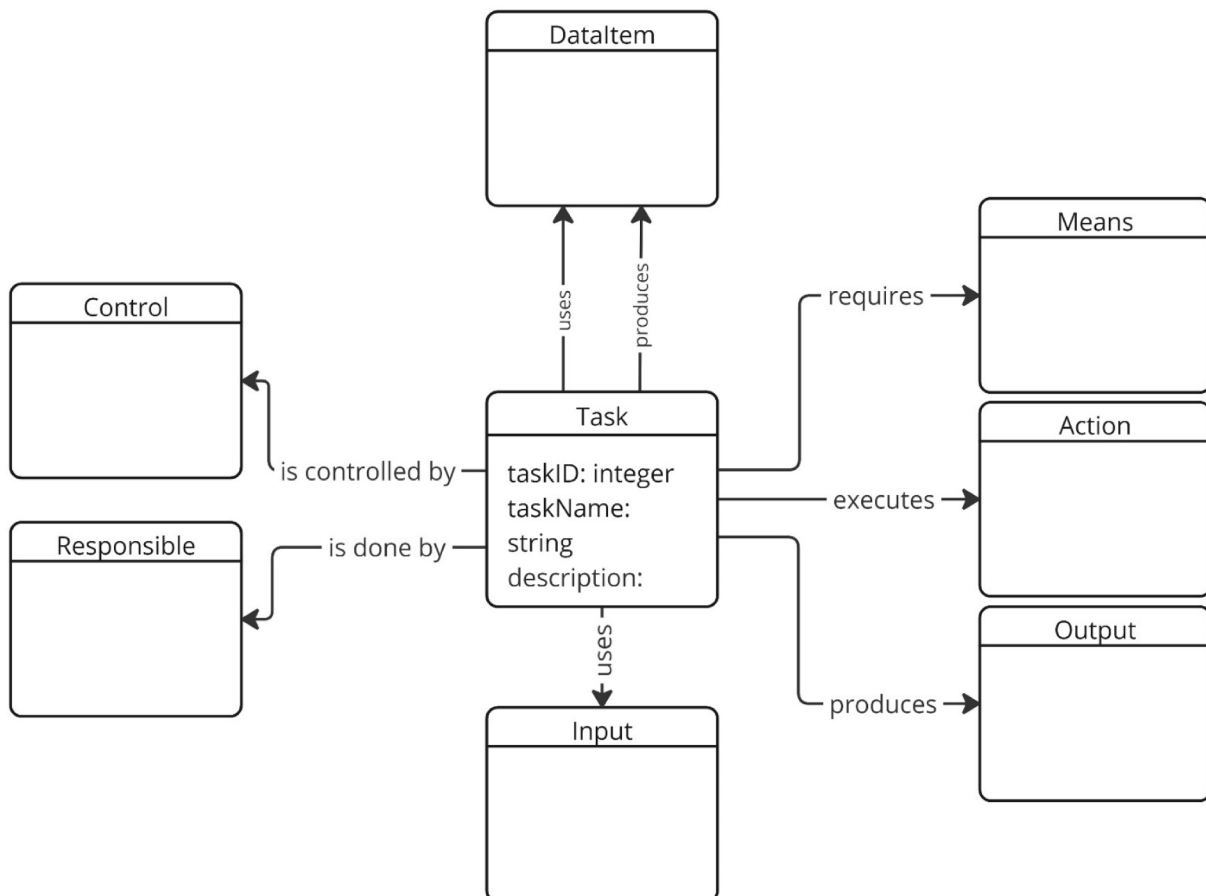


Fig. 2. Main classes in the ontology and their relationships.

Table 1

Class hierarchy of farm Tasks described in the ontology, organized in five levels.

Level 1	Level 2	Level 3	Level 4	Level 5
Planning	Financial_planning Land_use_planning	To be developed To be developed		
Operational/ tactical_management	Financial_management	To be developed		
	Production_management	Animal_management	Culling Animal_Health_Management Milking Nutrition/Feeding Purchasing Reproduction Selling Shearing Welfare Crop_management	Mating
		Soil/plant_management		Crop_Comercialization Crop_Fertilization_Management Harvesting_Crop Crop_Irrigation_Management Pest_Management_Crop Planning_Cropping Sowing_Crop
			Pasture_management	Pasture_Fertilization_Management Grazing_Management Harvesting_Pasture Pasture_Irrigation_Management Pest_Management_Pasture Sowing_Pasture
			Soil_health_management Soil/plant_Measuring R&M_Energy_Generation R&M_Energy_Storage Maintain_Milking_Shed Monitoring_Fences Repair_Fences Maintain_Machinery Monitoring_Machinery Repair_Machinery Scheduling_Machine_R&M Maintain_Farm_Shed Monitor_Farm_Shed Monitor_Milking_Shed Repair_Milking_Shed Repair_Farm_Shed	
	Repair_and_maintenance	R&M_Energy R&M_Milking_Shed R&M_Fences R&M_Machinery R&M_Farm_Sheds		
	Staff_management	Manage_Staff_H&S Manage_Staff_Payroll Roster_Management Staff_evaluation Staff_hiring		
	Assurance_reporting	Staff_training_and_development Workload_Manag.		
	Regulatory_reporting	Farm_Assurance Submitting_Reports_on_Farm_Assurance Animal_records Submitting_Reports_on_Animal_records Biosecurity Submitting_Reports_on_Biosecurity Environmental Submitting_Reports_on_Environmental Financial Submitting_Reports_on_Financial Health_and_safety Submitting_Reports_on_Health_and_safety		
	Supply_chain_reporting	Processors_requirements Submitting_Reports_on_Processors_requirements		

- h. Requires: means or mechanisms required by each task.
i. is_controlled_by: controls regulating each task.
j. is_done_by: person responsible for each task

2. All Actions
3. All Controls
4. All Means
5. All Entities

The full ontology diagram is depicted in Fig. 3. Also, the ontology is

freely available at the webprotege website, where it can be inspected and downloaded in form of an owl or rdf file from: <https://webprotege.stanford.edu/#projects/45b96c6f-d720-4223-a6f7-6d3915a90a49/edits/Classes>.

3.3. Network analysis

The network analysis shows a high level of interconnectivity between tasks, meaning that a large proportion of them require the same

Table 2
Number of classes and individuals on the main classes of the ontology. notice that individuals can have multiple inheritance, so the overall count does not correspond to the sum.

Main Classes	Number of subclasses	Number of individuals
Task	84	178
DataItem	20	330
Means	11	239
Action	47	180
Output		44
Input		45
Control	5	26
Entity	16	84
Total	174	661

data items, or that one task generates data required as data input for another task. This is illustrated in Fig. 4, which contains 121 nodes representing tasks for which linkages have been identified, with a total of 464 linkages. There were 96 data items linking at least 2 tasks, and up to 16 tasks.

A series of network metrics were calculated following Luthe and Wyss (2016). Among those individuals that show linkages, the average centrality degree (betweenness) was 3.05 (ranging from 1 to 12) for tasks and 3.08 for data items (ranging from 1 to 16). The average degree (number of linkages per node) for tasks was 7.67 (ranging from 1 to 26). The tasks with the highest degree related to financial budgeting, planning of animal sales and purchases, animal treatment recommendations and developing cropping policies. The density of the network (number of links in relations to the maximum possible number of links) as a whole

was 6.4 %, but there was a core of 15 tasks with a density of 100 %. The latter meaning that all the 15 tasks are linked to each other, these are quite diverse, sharing a variety of data items.

3.3.1. Clustering

We applied the Louvain method to identify task clusters. Thirteen clusters were found, containing between 2 and 21 tasks (Table 3). The modularity index for the network was 0.57, indicating a rather high tendency to form groups. The average clustering coefficient was 0.68, which we interpret as high, indicating the need for high volumes of information flows. Fig. 5 illustrate the formation of clusters in the network. It can be noticed that, even though there seems to be clear groups, there are also many dependencies between clusters. This indicates that it would be problematic to work with a cluster in isolation.

4. Discussion

4.1. The nature of the task

It has been postulated that the main barriers to the uptake of digital technologies in farming are socio-ethical, rather than technical (Shepherd et al., 2020). We believe that, even though problems do manifest as socio-technical barriers, there are root causes related to the very nature of farm digitalization. Our analysis seems to indicate that many of the difficulties experienced by farmers may originate in the multitude of disparate tasks (or functions) in farming systems, and the high degree of interdependency between them. The interdependencies between tasks will necessarily translate into interdependencies between digital technologies meant to support such tasks. Therefore, developing solutions

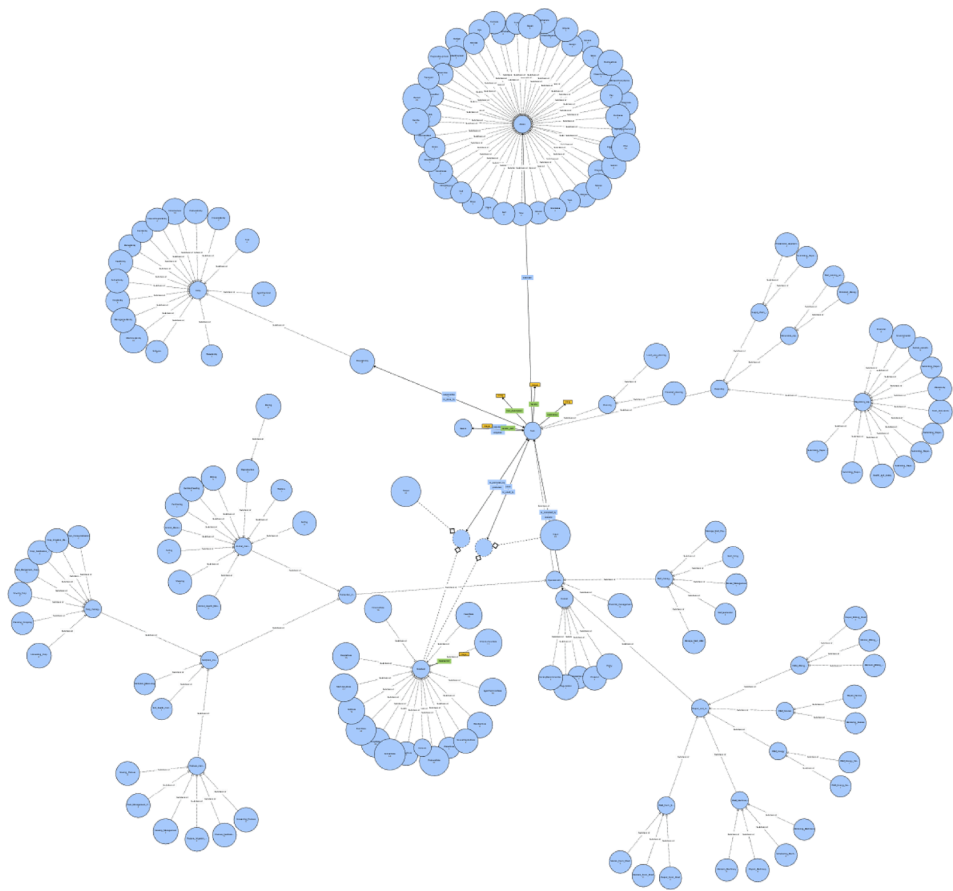


Fig. 3. Representation of all classes in the ontology and their interrelationships (done using <http://vowl.visualdataweb.org/>). Each circle represents a class (see Table 1) and the lines represent the relationships between each pair of classes (Fig. 2 shows the relationships among the main classes). This image is just an illustration, a high-definition version of this figure can be found in <https://doi.org/10.6084/m9.figshare.23615454>.

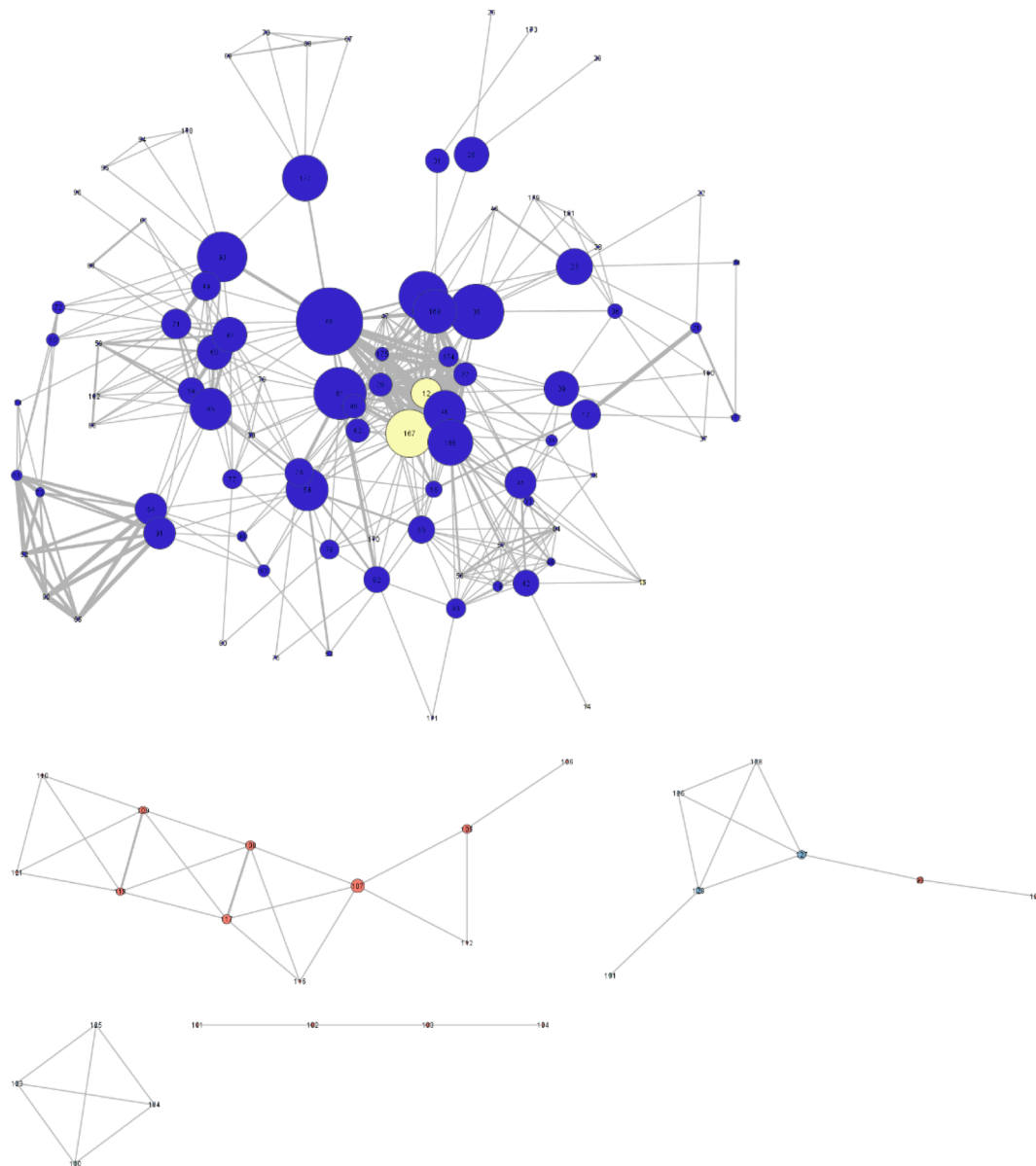


Fig. 4. Network diagram, where each node represents a farm task and links indicate that two tasks share data items. The size of the node is proportional to the centrality of the task (measured as betweenness, the extent to which a certain vertex lies on the shortest paths between other vertices, Hansen et al., 2010), the width of the links is proportional to the number of data items shared by two nodes, and the colours represent different classes of tasks (e.g., blue indicate “Production Management” tasks, and red “Repair and Maintenance”). Done with Visone (Brandes and Wagner, 2004).

for single tasks, or small groups of tasks, would inevitably cause the often cited problems of incompatibilities, lack of interoperability, data double handling and duplication, among other problems (Eastwood et al., 2023).

After our cataloguing and analysis of farm tasks, we contend that developing coherent digital solutions for farm systems that avoid these problems will require much more deliberate coordination and cooperation between developers and technology suppliers. Here we use the definitions offered by Castañer and Oliveira (2020): “*Coordination refers to the joint determination of common goals, while cooperation refers to the implementation of those goals*”. These authors conclude that cooperation follows coordination. For the former, they distinguish specific attitudes (“*willingness to work toward the achievement of an agreed-on common goal*”), behaviours (“*actions undertaken by the partners to achieve the collectively envisioned goal*”) and outcomes (e.g., “*the degree to which an agreed-on common goal is attained*”). Clearly, the coordination and cooperation that is required, we postulate, does not correspond with

what we see today in the agri-tech sector, with commercial actors working mostly independently. Solutions currently being proposed, such as data standards (DataLinker, 2023), are necessary but not nearly enough. There are simply too many data dependencies between farm tasks.

This does not mean, of course, that developers must work together on every single detail of their solutions. The concept of modularity comes in handy here. In technological design, the concept of modularity relies on the premise that “*by breaking up a complex system into discrete pieces—which can then communicate with one another only through standardized interfaces within a standardized architecture—one can eliminate what would otherwise be an unmanageable spaghetti tangle of systemic interconnections*” (Langlois, 2002). When referring to complex systems, Langlois uses the classic definition by Simon (1962): “*one made up of a large number of parts that interact in a non simple way. In such systems, the whole is more than the sum of the parts, at least in the important pragmatic sense that, given the properties of the parts and the laws of their interaction, it is not a trivial*

Table 3
Task clusters.

Cluster	Tasks related to	Number of Task	Number of Links
1	Repair_and_maintenance, Staff_management	7	9
2	Repair_and_maintenance	4	3
3	Repair_and_maintenance	11	20
4	Staff_management, Assurance_reporting	4	6
5	Production_management	9	19
6	Financial_management, Production_management	14	91
7	Production_management	15	27
8	Production_management	2	1
9	Production_management	5	10
10	Production_management	21	68
11	Financial_management, Production_management	16	50
12	Production_management	3	2
13	Production_management	10	29
Total		121	335

matter to infer the properties of the whole”. So, indeed, digital systems can be broken down into modules with various degrees of interdependence (Kostakis, 2019), the idea being to modularize the system in a way that minimizes dependencies or at least making dependencies clear so that they can be actively addresses. However, even though our analysis shows a level of modularity in the network, it also shows multiple dependencies between modules. Therefore, according to our analysis, there is no way to partition farm tasks into modules so that each one can be developed in isolation, even if data standards were adopted. This would be akin to expect that standards would make independent work on wheels, bodies and engines result in a well-functioning car design. Just like cars are designed like units (although they may use generic parts), all the techno-digital components of a modern farm also need to be seen as a unit or system. We call such system “the digital layer”, which would include all the software, communication and hardware elements of a farm. An integrative design approach is needed, targeting whole systems simultaneously, instead of parts (Kulak et al., 2016; Lovins, 2010). According to Cross et al. (2015) an integrated approach should “emphasize designing all the systems together, rather than sequentially, in order to ensure they work in harmony”.

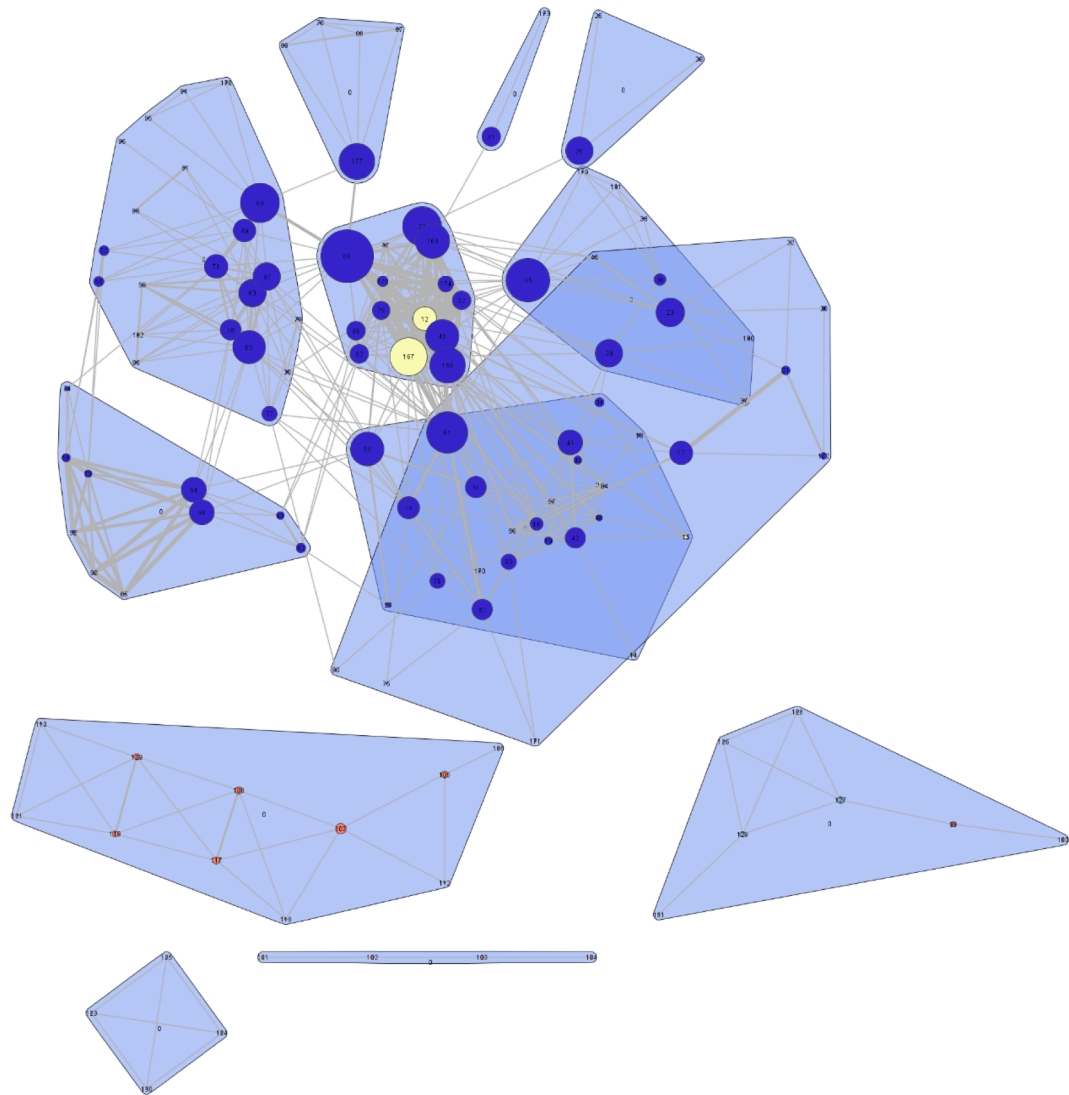


Fig. 5. Network diagram, where each node represents a farm task and links indicate that tasks share data items. The size of the node is proportional to the centrality of the task (measured as betweenness, the extent to which a certain vertex lies on the shortest paths between other vertices, Hansen et al., 2010), the width of the links is proportional to the number of data items shared by two nodes, and the colours represent different classes of tasks (e.g., blue indicate “Production Management” tasks, and red “Repair and Maintenance”). The grouping in blue represents clustering using the Louvain clustering method. Done with Visone (Brandes and Wagner, 2004).

4.2. Required organizational structures

As established above, to be truly effective for farmers, the digital layer should be seen as a system. To such end, in our analysis we represented the problem as a network of tasks, where nodes represent farm tasks and linkages represent data dependencies. This infers that we can also see the ‘digital layer’ as a network of technological components. Furthermore, we can see the development of the digital layer as a “collective action problem”. Collective action means members of a group working together to achieve some common objective. Collective action problems arise from the fact that, despite sharing common interests, members also have conflicting interests (Dowding, 2013). Collaborative networks can be set up to organize collective action, but, according to Bodin (2017) the structure of such networks needs to fit the nature of the collective action problem. Bodin describes the problems arising when interdependent ecological components are managed by different actors who do not coordinate their activities, and argues that ecosystems should be managed as systems and not as sets of isolated components. He was referring to environmental and ecosystem management problems, but we believe his insights may also apply to our digital layer problem. Nevertheless, as stated in the previous section, the techno-digital components in farming are not being regarded as a system, with technology actors not collaborating effectively, if at all, and farmers experiencing an “spaghetti tangle” situation (Langlois, 2002). This is illustrated in Fig. 6, based on the New Zealand situation.

The lack of coordination in the digital technology supply is a barrier for farmers. Indeed, as stated by Amiri-Zarandi et al. (2022), “a major challenge in smart farming information processing is consistency and compatibility among the utilized technologies, procedures, and protocols”. What would ideally be needed is some sort of **integration system**, to potentially enable any farm components with digital elements (**software and hardware**) to exchange information with any other. Clearly the *laissez-faire* approach that is now being followed is not working well.

A pertinent question to ask then would be what sort of organizational structure would need to be in place for the level of cooperation and coordination that is needed? Some type of **entity** would need to design

and support the digital layer system. So, the software, hardware, the integration system altogether, constitute the digital layer (Fig. 7). This would be a rather centralized system, with a main integration system controlling the exchanges. Another alternative would be a decentralized system, web3.0 style, where a sort of market is established that puts monetary value on digital assets (Romera et al., 2024). After our analysis however, we believe that the required level of coordination needed could not be achieved without steering by a **central actor**.

Some of the functions of such **central actor** would be:

1. **Design generic** digital layers
2. **Guide** the tech developers by identifying gaps to be filled.
3. Holistic **design** of the digital layer for specific farms. This could involve using new components and/or connecting components already existing on the farm.
4. **Build** digital layers. This would mean building the integration system for a farm/client, and establishing the connection between the components.
5. **Support** the farmers (the client) in the setup and operation of the digital layer. This could include software updates, cloud services, cybersecurity, and support when new components or functions need to be incorporated to the layer.
6. **Develop** and maintain the integration platform.
7. **Coordinate** API developments for integration of third-party software and data into the digital layer system.
8. Develop and enforce data **standards**.
9. Coordinate **new developments**, which could be achieved through a variety of modes of governance, including more hierarchical, centralized structures, or more open-source, decentralized structures (Williamson, 2005).

Having established the case for a central actor, it is obvious that this would not necessarily be what will happen, as this type of collective solutions are notoriously difficult to achieve in practice (Dowding, 2013). Other than **non-coordination**, with the continuation of the current situation and with suboptimal realization of the potential of

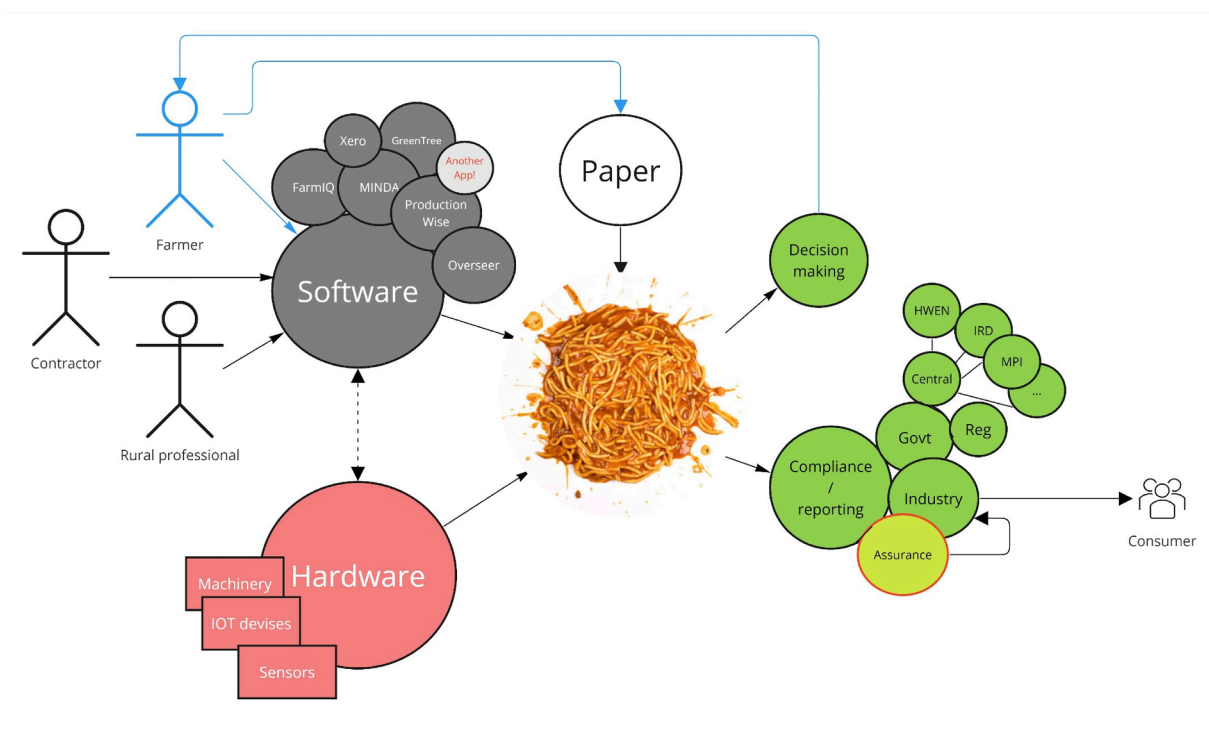


Fig. 6. Spaghetti tangle situation on farm digitalization (New Zealand illustration), arrows indicate data flows, but necessarily digital.

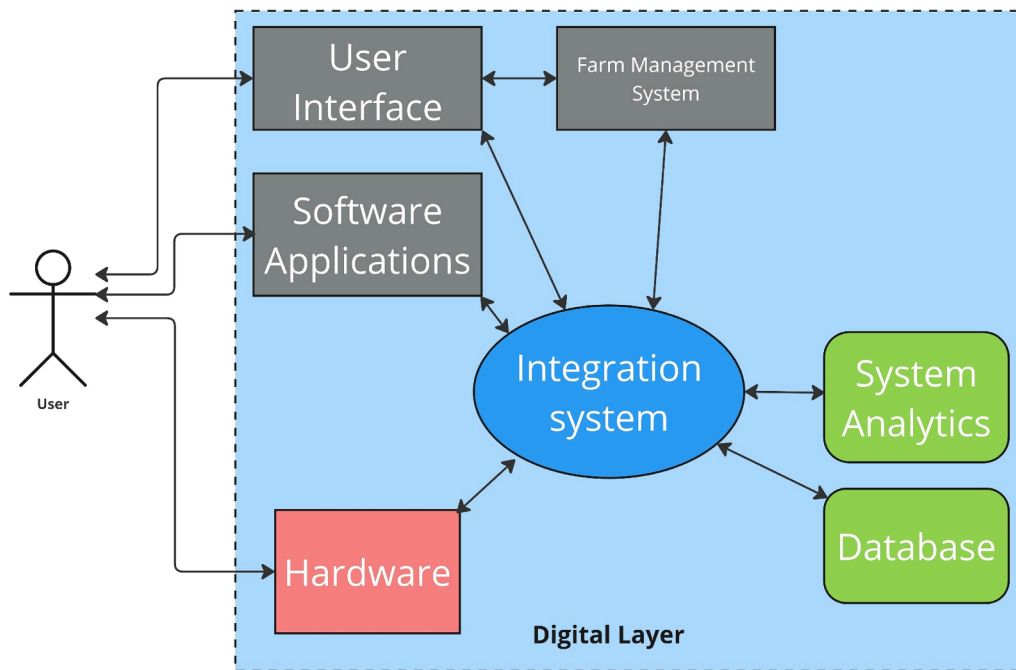


Fig. 7. Envisioned situation in which all components of the farm digital layer interact and exchange data (indirectly) through a dedicated Integration System.

digital technologies, we envision five possible future scenarios for the central actor, or hybrids between them:

1. **Commercially owned:** a firm as the central owner. This could be a privately owned firm, a government owned company (for profit user-pay) or public-private partnership.
2. **User owned:** it could be a large food company or a farmer's organization. In the latter case, for example, it could function as a consumer cooperative, or as a levy organization.
3. **Developer owned:** a community of developers like Linux or R, with a central coordination entity. It could be organized as a worker cooperative, or Commons-based peer production (Kostakis, 2019).
4. **Public service:** a state-owned organization supporting coordination among developers (commercial or state employees). This enterprise would deliver a service to farmer, in the same way as, in many countries, the state funds extension services and support to farmer at no direct cost to them.

Further research is necessary to analyse the implications of each of these non-mutually exclusive scenarios.

4.3. Components of the digital layer

In our definition, the digital layer as a system consists of cyber and cyber-physical sub-systems co-ordinated by the integration system, which facilitates co-operation among digital technologies for effective management and operation of the farm. That is, supporting the fulfilment of farm tasks (see section 3.1). Fig. 7 shows the main components of the layer and their interrelationships, below we briefly describe each component. This proposed architecture for the digital layer is a generic one that incorporates both the IoT reference architecture (ISO/IEC 30141:2018, International Organization for Standardization, 2018) and the digital twin architecture (Aheleroff et al., 2021) components. Standards and references like these would be formally incorporated during the detailed design stage of the farm digital layer.

Hardware: hardware includes cyber-physical components such as sensors, IoT devices, drones, irrigators, tractors and other farm

implements, cloud servers, etc.

Farm Management System: is the farm main software application where basic functions for operation, management, planning, reporting, and record keeping are available for a farmer to use.

Software Application: includes any 3rd party software systems (web-based or OS based) that contribute to specialised areas of farm management/planning.

Database: includes the main data storage system for the farm digital layer, where farm's main data, shared data from 3rd party software applications, and incoming data from on-farm sensors are stored and managed.

System Analytics: an analytics platform (cloud or local server) where farm data processing for a variety of analytics and decision support purposes are performed. Notice that 3rd party components also perform their own analytics (e.g. edge computing).

Integration System: includes a data integration platform (for brokering data between layer components, as well as 3rd party systems through API), and an operating/application framework which binds all the digital layer components together to run the system effectively.

User Interface: the front-end dashboard system (ideally web-based) where the users can interact with the digital layer system, specifically the farm management system.

User: Those who interact with the digital layer system through the user interface, main users are farmers (farm manager, farm worker, etc), advanced admin users, external users given permission to access certain data (compliance agency, food processing company, accountants, rural professionals, etc) or certain functions (reporting, visualisations, etc).

4.4. Prototyping

Clearly, prototyping is a key element in design processes. In this section, we introduce an existing 'farm digital platform' – farmOS, and comment on some of its characteristics in relation to prototyping several aspects of the farm digital layer concept.

Main characteristics of farmOS:

farmOS is an open-source farm management software platform designed to streamline and simplify the management of farm operations

(farmOS, 2023). It provides farmers with a comprehensive set of tools for planning, tracking, and analysing various aspects of their farm activities. Whether it is a small-scale farmer or managing a large organisation, farmOS offers a range of features and functionalities to support farm management tasks.

farmOS is built on the principle of open-source software, which means it is freely available to download, use and modify. This allows farmers to customize and adopt the software to suit their specific requirements. The platform aims to empower farmers with data-driven insights and efficient farm management practices, ultimately improving productivity, sustainability, and profitability.

This platform encompasses several of the key components that we identify as the digital layer concept, as depicted in Fig. 7, making it very useful for our prototyping process. Such components include farm management functionality, database, integration system, user interfaces, and framework for developing software tools analytics.

Key features and benefits of farmOS:

Farm management, planning and reporting tool: farmOS provides a comprehensive set of tools for managing farm operations, planning, reporting, and record keeping. It utilises a structured database schema to store and organise data related to the different aspects of a farm. Input, query, and management of farm data are powered through relational database management systems such as PostgreSQL, MySQL, or MariaDB.

Modular Architecture, Integration and Extensibility: farmOS is built on modular architecture based on the Drupal content management system. Drupal allows users to select and install different modules (applications) into the farmOS platform to add customised functionality. These modules are either developed by the community or custom built, allowing flexibility in utilising a wide range of digital functions serving certain farm needs. Drupal-based custom modules can be developed using PHP programming languages and added to the system.

farmOS provides a range of integration options to connect to other systems and services for data exchange through RESTful APIs (farmOS 1.x) and JSON:API (farmOS 2.x). Every defined farm entity (asset, log, sensor, animal) is provided with an API end point, which data can flow into from external services, or IoT sensors, and stored against the registered asset in the database.

Data Visualization and Analysis: Farmers can generate reports, graphs, and charts to visualize their farm data. This allows for in-depth analysis of trends, historical records and performance metrics and make data-driven decisions.

Data Security and Privacy: farmOS uses industry standard security practices to protect data, including user authentication mechanisms, role-based access control, deployability on a secure infrastructure utilising HTTPS, and SSL encryption.

In sum, these capabilities, together with the open-source nature of farmOS has been a good enabler for us to explore the farm digital layer system design through prototyping.

5. Conclusions

As previously stated, and noted by many authors, digital technologies are yet to fully deliver on their promises in agriculture. Our working hypothesis was that, at least in part, the problem lays in the piecemeal way in which digital technologies are being developed and delivered to farmers. In this study we took an engineering approach to the conceptual design of an integrated digital system to support farming. We call such system the 'digital layer', which encompasses all the software and digitally supported hardware on a farm, as well as integration platforms and cloud services.

Our starting point was the cataloguing and high-level description of tasks on a generic dairy farm, including their inputs, outputs, controls, and the means that each task requires. The information was then organized into an ontology of farm tasks. This systematic way of organizing the knowledge allowed us to conduct a network analysis to explore the

interdependencies between farm tasks, specifically in terms of data inputs and outputs. Our findings revealed a high level of interdependency between farm tasks. This led us to conclude that the farm digital layer needs to be integrally designed (in the sense discussed by Lovins, 2010), delivered, and supported. This would require a much higher degree of coordination than we currently observe in the agri-tech sector, and possibly imply the need for central coordination entities.

CRedit authorship contribution statement

A.J. Romera: Conceptualization, Data acquisition, Analysis, Writing. **M. Sharifi:** Conceptualization, Data acquisition, Analysis, Writing. **S. Charters:** Conceptualization, Writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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